Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. You prompt ChatGPT with: "Translate 'Good morning' into French." What transformer component is most active during this phase? 1 / 1 point

- (•) Encoder-decoder
- ( ) Encoder only
- ( ) Decoder only
- ( ) Masking layer

Nice work

Correct: In encoder-decoder models, the encoder processes the source language while the decoder uses that context to produce the translation step-by-step.

2. Below is a simplified attention map of a sentence processed by a transformer: 1 / 1 point



What can you infer from the map?

- ( ) "Supper" is unrelated to "cook"
- (•) The model considers "spatula" and "supper" important to predict "cook"
- ( ) All words have equal importance
- ( ) "Cook" influences "spatula" and "supper"

Nice work

Correct: This map shows that the model has assigned higher weights to "spatula" and "supper" when processing "cook," which means these words provide meaningful context.

3. A hospital chatbot is designed to fill in missing details in patient instructions. For instance: "Take your medication after ___ five days." What type of model architecture is this task best suited for? 1 / 1 point

- ( ) Summarization model
- ( ) Decoder-only language model
- ( ) Named entity recognition model
- (•) Masked language model

Nice work

Correct: Masked language models like BERT are trained to fill in missing tokens by analyzing both the left and right context. This makes them ideal for tasks that require predicting a word in the middle of a sentence.

4. What is the role of the encoder in a transformer model? 1 / 1 point

- (•) Create contextual embeddings for input text
- ( ) Filter out stopwords
- ( ) Translate input to output
- ( ) Generate the next word

Nice work

Correct: The encoder processes the input sequence to create contextualized representations that are used for downstream tasks.

5. Scenario: A model fills in the blank: "He picked up his ___ to cook supper." What kind of model is most likely being used? 1 / 1 point

- ( ) Translation model
- ( ) Sentiment analyzer
- ( ) Named Entity Recognizer
- (•) Masked language model

Nice work

Correct: This is a classic example of masked language modeling, where a word is hidden and the model is trained to infer it using context.

6. What does a language model typically predict? 1 / 1 point

- ( ) The meaning of a sentence
- ( ) The next paragraph in a document
- ( ) The grammatical structure of a sentence
- (•) The next word in a sequence

Nice work

Correct: Language models are designed to predict the most likely next word based on a given sequence of words. This prediction is grounded in probability, using context from previous words.

7. Which components contribute to the final output of a transformer model? (Select all that apply) 1 / 1 point

- [x] Attention weights

Nice work

This is one of the correct answers: Attention weights determine how much importance is given to each token.

- [x] Decoder predictions

Nice work

This is one of the correct answers: The decoder converts internal representations into output tokens.

- [x] Encoder representation

Nice work

This is one of the correct answers: The encoder produces contextual embeddings that guide downstream prediction.

- [ ] Dropout rate

8. Transformers can be fine-tuned for a wide variety of NLP tasks. 1 / 1 point

- (•) True
- ( ) False

Nice work

Correct: Fine-tuning allows the base transformer model to adapt to specific downstream tasks like named entity recognition, sentiment analysis, or question answering.

9. Which of the following techniques are used to generate text from language models? (Select all that apply) 1 / 1 point

- [ ] Singular Value Decomposition (SVD)
- [x] Greedy decoding

Nice work

This is one of the correct answers: Greedy decoding selects the most probable word at each step, producing fast but sometimes repetitive text.

- [x] Beam search

Nice work

This is one of the correct answers: Beam search keeps multiple hypotheses alive, improving fluency and coherence.

- [x] Sampling

Nice work

This is one of the correct answers: Sampling introduces randomness, helping generate diverse outputs.

10. What task does a masked language model perform? 1 / 1 point

- (•) Fills in a missing word within a sentence
- ( ) Predicts the next sentence
- ( ) Clusters documents by topic
- ( ) Generates part-of-speech tags

Nice work

Correct: Masked language models (e.g., BERT) are trained to predict missing tokens in a sentence by learning from the surrounding context.

11. Which statement about beam search is true? 1 / 1 point

- ( ) It always selects the shortest sentence
- ( ) It is used only during model training
- ( ) It randomly chooses outputs
- (•) It tracks the k most likely sequences during generation

Nice work

Correct: Beam search maintains multiple high-probability sequences (typically top-k) to improve generation quality.

12. Transformers require labeled data during pretraining. 1 / 1 point

- ( ) True
- (•) False

Nice work

Correct: Labeled data is only required during fine-tuning, not for the initial training phase.

13. What enables transformers to be used for zero-shot tasks? 1 / 1 point

- (•) Pretraining on large corpora
- ( ) Use of CNN layers
- ( ) Manually defined rule sets
- ( ) Fine-tuning on small datasets

Nice work

Correct: Pretraining on massive text corpora equips transformers with broad general knowledge that enables zero-shot generalization.

14. Which of the following are true about self-attention in transformers? (Select all that apply) 1 / 1 point

- [x] It allows a word to attend to all others in a sentence

Nice work

This is one of the correct answers: Unlike RNNs, transformers use global attention—every word can attend to every other word.

- [x] It increases model parallelism

Nice work

This is one of the correct answers: Because self-attention does not require sequential computation, it enables better parallelism.

- [x] It helps identify relevant context words

Nice work

This is one of the correct answers: Self-attention allows the model to assign importance to different words in the input, enhancing context sensitivity.

- [ ] It replaces activation functions

15. Which of the following are true about transformer-based large language models? (Select all that apply) 1 / 1 point

- [x] They can generate text using beam search

Nice work

This is one of the correct answers: Beam search helps generate coherent and high-quality text by exploring multiple likely word sequences in parallel.

- [ ] They require recurrent layers to track history
- [ ] They use labeled data during pretraining
- [x] They learn internal representations of language

Nice work

This is one of the correct answers: Transformer models learn dense embeddings that represent both meaning and syntactic context, which are used to improve downstream tasks.